HMO Licence Retention and Rental Supply in St Andrews

Longitudinal administrative-data construction, property matching, regression analysis, and historical web scraping

Project objective

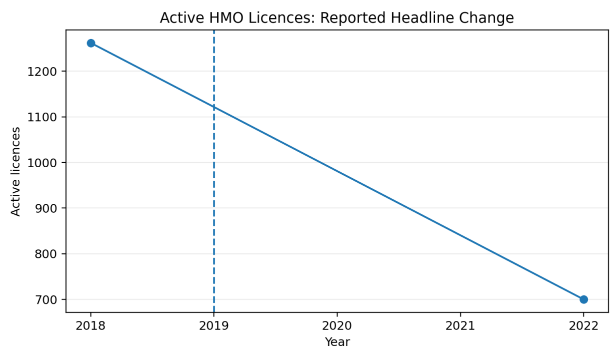
Measure changes in the supply and retention of HMO-licensed properties surrounding Fife Council's 2019 freeze on new HMO licences, and explore historical rental-price evidence from archived property listings.

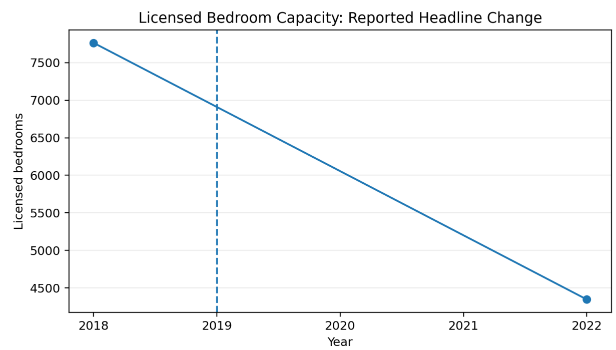
Data and analytical workflow

HMO register: Converted Fife Council records from PDF to Excel, then used R to clean, standardize, deduplicate, and match property-level records across 2004-2022. The resulting panel supported annual licence counts, licensed-bedroom estimates, and retention models.
Historical rental listings: Used the Wayback Machine, Selenium, Beautiful Soup, and Python to collect and clean archived rental listings from 2012-2020. The analysis included exploratory rent distributions and an Ayton House case study.

Key findings

• Constructed a cleaned longitudinal database containing **4,918 HMO licence records**.
• Active HMO licences fell from **1,262 in 2018 to 700 in 2022** (44.5%).
• Licensed bedroom capacity fell from **7,764 to 4,347** over the same period (44.0%).
• Larger properties had materially higher observed licence-retention rates than three-occupant properties.
• Conservation-area status was not statistically significant in the main or alternate-period model.
• The historical scraper collected about **1,400 raw listings**, cleaned to **478 relevant St Andrews records**.





Interpretation and limitations

The results document a substantial contraction in the observed stock of HMO licences and licensed bedroom capacity surrounding the policy period. The design is observational and does not isolate the freeze from all concurrent changes. Archived rental listings are incomplete, nonrandom, and uneven across years; those findings are exploratory. The Ayton House analysis is retained in the repository and full report, but its original input file is no longer available for rerunning.

Tools: R, Python, Selenium, Beautiful Soup, pandas, Excel, regression modeling, record linkage, and data visualization.